

Towards Robust Instrument Segmentation with Data Augmentation

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Abstract. Accurate and robust segmentation of surgical instruments is critical for effective analysis of surgery videos. In this paper, we present our solution to the MICCAI 2024 SegSTRONG-C challenge, focusing on enhancing segmentation performance under challenging conditions. We introduce a data augmentation technique that simulates the presence of surgical smoke. Combined with AutoAugment, we train a U-Net model to automatically segment instruments and the model achieves mean Dice Similarity Coefficient scores of 0.87, 0.43, and 0.92 across the domains of smoke, low brightness, and blood, respectively. The code is publicly available at <https://github.com/ProjectNeura/SegSTRONGC>.

1 Introduction

The SegSTRONG-C challenge [3] intends to draw attention to the robustness of models, which are trained on regular datasets, dealing with non-adversarial corruptions. The main obstacle to the desired robustness is the damage to tool edges caused by the non-adversarial noise, which largely determines the upper limit of the tasks, and the fact that target domains are not given. To overcome these issues, augmentation is necessary. In the baseline results, we see state-of-the-art results from the combination of UNet and AutoAugment in all domains [3]. Yet, we believe room for improvement still exists in domains of blood and low brightness. Based on the promising potential, we propose an augmentation method to simulate the broken edges to increase the capability of the model to recover the mask boundaries.

2 Methods

Our approach is implemented using the baseline model included in the official framework CaRTS [4] but fine-tuned with custom augmentations based on Albumentations [1], which is a library for fundamental image augmentation. According to the baseline results [3], we chose UNet as the network, and the checkpoint trained with UNet and AutoAugment as the pre-trained weights. CaRTS defines a workflow for model training and inference with single GPU support. By alternating the configuration, we were able to fine-tune the model with ease.

2.1 UNet

UNet is a U-shaped convolutional neural network designed for image segmentation [6]. Its capability in biomedical image processing has been proven for long. It consists of 4 down-size layers and 4 up-size layers, followed by a convolution layer as the output layer, where all layers are convolutional except the up-size ones. The kernel size of the down-size layers and the output layer are 3 and 1 respectively. Each up-size layer is made of a bilinear up-sampler with a scale factor of 2 and a double convolutional module, which contains convolution, batch normalization, and ReLU.

2.2 Optimization

We fine-tune the pre-trained model on the augmented dataset, which is based on the original training dataset, using Stochastic gradient descent [5] as the optimizer with a learning rate of $3e-4$. The training process ran on a single Nvidia RTX 4090 (24GB) for 40 epochs.

2.3 Custom Augmentations

Smoke Generation Transformations, which refer to rotation and crop, are applied to a high-resolution texture to introduce randomness. These transformations are implemented with the help of Albumentations [1]. The randomized smoke image is then merged on top of the original image with an opacity of 1. In addition to the smoke texture described above, another noticeable pattern observed from the example image given on the challenge website is the effect of diffuse. Therefore, we generate a circle flare in which the border is blurred.



Fig. 1. Comparison between the Original Image, the Augmented Image, and the Official Example

Other Augmentations We have tried augmentations for blood and low brightness, but our implementation does not seem to be better than AutoAugment [2].

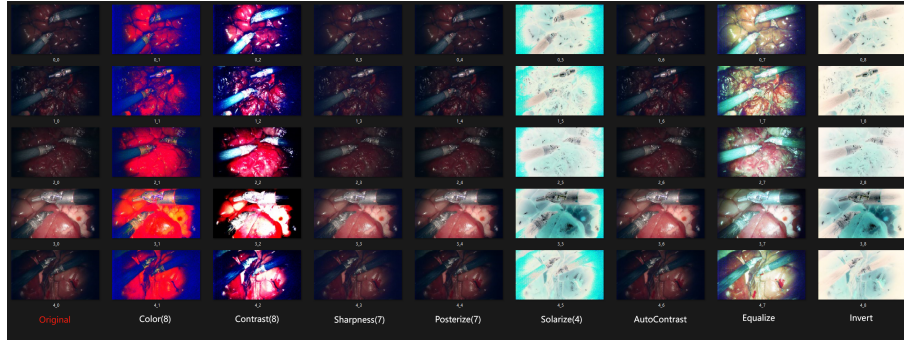


Fig. 2. AutoAugment

3 Results

3.1 Evaluation Method

We evaluate and compare our model with the original one using the Dice Similarity Coefficient and Normalized Surface Distance on the validation set that is augmented from the provided validation set. Besides the smoke augmentation described previously, a low-brightness augmentation by Albumentations [1] and a blood augmentation (random cell growth) are applied.



Fig. 3. Model's Prediction on an Augmented Image

3.2 Quantitative Results

Compared to the baseline model, our approach achieved state-of-the-art results in all domains by both metrics. The limited improvement in the low-brightness scenario is likely due to the loss of information when the images are compressed to a tiny range of brightness.

Metric	UNet+AutoAugment	Ours
Mean DSC (Smoke)	0.1872	0.8666
Mean NSD (Smoke)	0.1687	0.7691
Mean DSC (Low Brightness)	0.2405	0.4279
Mean NSD (Low Brightness)	0.2094	0.3889
Mean DSC (Blood)	0.9061	0.9201
Mean NSD (Blood)	0.8780	0.8959

Table 1. Comparison of DSC and NSD Results

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